MTH636 – Game Theory

Lecture 20: Nash Equilibrium, SPE & Backward Induction

1 Extensive Form Games with Chance – Recap

Recall that a general extensive form game (possibly with a chance player) is described by the tuple:

$$\Gamma = \langle N,\ V,\ E,\ n_0,\ (V_i)_{i\in N\cup\{0\}},\ (p_n)_{n\in V_0},\ (u_i)_{i\in N}\rangle$$

1.1 Example: Sequential Battle of the Sexes with Chance

Consider the following game tree. Nature (player 0) moves first with probability $\frac{1}{2}$–$\frac{1}{2}$, choosing between two "types": Man (M) or Woman (W) plays first. Subsequently the other player responds.

Game Tree Description.

- The root (chance node) splits with probability $\frac{1}{2}$ to the left subtree (Man moves first) and $\frac{1}{2}$ to the right subtree (Woman moves first).
- **Left subtree** (Man moves): Man chooses F or D.
  - If Man plays F: Woman chooses F (payoff $(2,1)$) or D (payoff $(0,0)$).
  - If Man plays D: Woman chooses F (payoff $(0,1)$) or D (payoff $(1,2)$).
- **Right subtree** (Woman moves): Woman chooses F or D.
  - If Woman plays F: Man chooses F (payoff $(2,1)$) or D (payoff $(0,0)$).
  - If Woman plays D: Man chooses F (payoff $(0,0)$) or D (payoff $(1,2)$).

Consider the strategy profile $\Big((F,F,D),\ (D,F,D)\Big)$, i.e., Man plays F in the left subtree and plays (F,D) in the right subtree; Woman plays (D,F,D).

Expected utilities:

$$U_M\Big((F,F,D),(D,F,D)\Big) = \tfrac{1}{2}\cdot x_0 + \tfrac{1}{2}\cdot x_1 = \tfrac{1}{2}$$
$$U_W\Big((F,F,D),(D,F,D)\Big) = \tfrac{1}{2}\cdot x_0 + \tfrac{1}{2}\cdot x_2 = 1$$

2 Nash Equilibrium in Extensive Form Games

Definition: Nash Equilibrium (Extensive Form)

A strategy profile s^* is a **Nash Equilibrium (NE)** if, for all $i \in N$ and all $s_i \in S_i$:

$$u_i(s^*) \geq u_i(s_i,\ s^*_{-i})$$

Given strategy sets $S_1, \ldots, S_n$, the corresponding **strategic (normal) form** game is:

$$\langle N,\ (S_i)_{i\in N},\ (\hat{u}_i)_{i\in N}\rangle$$

Theorem: NE Equivalence (Result K-1)

For an extensive form game Γ, a strategy profile s^* is a Nash Equilibrium of Γ **if and only if** s^* is a Nash Equilibrium of the corresponding strategic form game.

3 Illustrative Example: NE vs. SPE

Consider the following extensive form game. Man (M) moves first, choosing F or D.

- If Man plays F: Woman (W) chooses F (payoff $(2,1)$) or D (payoff $(0,0)$).
- If Man plays D: Woman (W) chooses F (payoff $(0,0)$) or D (payoff $(0,2)$). (The underlined value $(0,2)$ denotes a Nash-relevant payoff.)

The corresponding normal form matrix (Man's strategies: F, D; Woman's strategies: (F,F), (F,D), (D,F), (D,D)) is:

	(F,F)	(F,D)	(D,F)	(D,D)
F	$(2,\ 1)$	$(2,\ 1)$	$(0,\ 0)$	$(0,\ 0)$
D	$(0,\ 0)$	$(1,\ 2)$	$(0,\ 0)$	$(1,\ 2)$

Analysing the Nash Equilibria and SPE status of various strategy profiles:

Strategy profile	NE?	SPE?
$\big(F,\ (D,F)\big)$	Not an NE	– (W has a true deviation)
$\big(F,\ (F,F)\big)$	NE	**not** SPE
$\big(F,\ (F,D)\big)$	NE	SPE
$\big(D,\ (D,D)\big)$	NE	**not** SPE
$\big(D,\ (F,D)\big)$	Not an NE	– (M has a true deviation)

Note

$\big(F,(F,F)\big)$ is an NE but **not** an SPE: at Man's node under D, Woman would prefer D over F but she is committed to (F,F), which is non-credible off the equilibrium path.
$\big(D,(D,D)\big)$ is an NE but **not** an SPE: if Man deviates to F, Woman's best response is F (payoff $1>0$), so threatening to play D after F is a non-credible threat.

4 Subgame Perfect Equilibrium (SPE)

Definition: Subgame Perfect Equilibrium

A strategy profile s^* is called a **Subgame Perfect Equilibrium (SPE)** if, for every subgame of the game, s^* restricted to that subgame is a Nash Equilibrium of that subgame.

Note

Every SPE is also a Nash Equilibrium of the whole game. The converse is not true in general.

5 Backward Induction Equilibrium

Backward induction is an algorithm for finding equilibria in finite extensive form games with perfect information. One works from the leaves of the game tree back to the root, at each node selecting the action that maximises the current player's payoff.

5.1 Example 1: Two-player Sequential Game

Original game tree. Man (M) moves first: F or D.

- F branch: Woman (W) chooses F (payoff $(2,1)$) or D (payoff $(0,0)$).
- D branch: Woman (W) chooses F (payoff $(0,0)$) or D (payoff $(1,3)$).

Backward induction step.

- At Woman's node after F: payoff $(2,1)$ vs. $(0,0) \Rightarrow$ Woman plays **F** (payoff $1 > 0$).
- At Woman's node after D: payoff $(0,0)$ vs. $(1,3) \Rightarrow$ Woman plays **D** (payoff $3 > 0$).

After resolving Woman's choices, Man faces: $F \to (2,1)$, $D \to (1,3)$. Man chooses **F** (payoff $2 > 1$).

$$\boxed{\text{Backward Induction Equilibrium: } \big(F,\ (F,D)\big)}$$

5.2 Example 2: Game with Chance Node and Multiple Subgames

Game tree. Player I moves first, choosing A, B, or C.

- A: Player II moves (node II), choosing l (payoff $(4,1)$) or g (payoff $(7,2)$).
- B: Player II moves (node II), choosing l' (payoff $(6,3)$) or g' (payoff $(4,0)$).
- C: Chance node splits $\frac{1}{4}/\frac{3}{4}$.
  - With prob. $\frac{1}{4}$: Player D moves, choosing L (payoff $(0,0)$) or R (payoff $(8,4)$).
  - With prob. $\frac{3}{4}$: Player I moves, choosing A' (payoff $(8,8)$) or b (payoff $(0,8)$).

Backward induction.

- Under A: Player II prefers g (payoff $2 > 1$). Resolved payoff: $(7, 2)$.
- Under B: Player II prefers l' (payoff $3 > 0$). Resolved payoff: $(6, 3)$.
- Under C, prob. $\frac{1}{4}$ branch: Player D prefers R (payoff $(8, 4)$). Resolved: $(8, 4)$.
- Under C, prob. $\frac{3}{4}$ branch: Player I prefers A' (payoff $8 = 8$). Resolved: $(8, 8)$.
- Expected payoff under C: $\frac{1}{4}(8, 4) + \frac{3}{4}(8, 8) = (8, 7)$.

Player I now chooses among $(7, 2)$, $(6, 3)$, $(8, 7)$: Player I prefers C (payoff 8).

After the second round of backward induction, Player I's node reduces to choosing $A \to (7, 2)$, $B \to (6, 3)$, $C \to (8, 7)$. Player I chooses **C**.

$\boxed{\big((C,\, a),\ (g,\, l',\, R)\big) \text{ is the Backward Induction Equilibrium.}}$